

Computer Networks — Network Layer & Routing

Complete Revision Notes (Hinglish)

UNIT: NETWORK LAYER

1. Network Layer

OSI Model ki 3rd layer.

Main Functions

- Routing
- Logical Addressing
- Packet Forwarding
- Internetworking
- Fragmentation
- Congestion Control

Devices

- Router
 - Layer 3 Switch
-

2. Network Layer Design Issues

(A) Store and Forward Packet Switching

Router pehle packet receive karta hai, store karta hai, fir next router ko bhejta hai.

Flow

Sender → Router → Router → Receiver

Advantage

- Reliable

Disadvantage

- Delay increase
-

(B) Connectionless Service

- No setup required
- Har packet independent
- Different routes possible

Example

IP Protocol

Advantage

- Fast

Disadvantage

- Packets out of order aa sakte hain
-

(C) Connection-Oriented Service

- Pehle connection setup
- Fir data transfer

Advantage

- Ordered delivery

Disadvantage

- Setup overhead
-

(D) Routing

Best path choose karna.

Goals

- Minimum delay
 - Less congestion
 - Efficient delivery
-

3. Logical Addressing

Different networks me communication ke liye logical address use hota hai.

IP Address

Device ka unique logical address.

Types

- IPv4
 - IPv6
-

4. Basic Internetworking

Different networks ko connect karna.

Devices Used

- Router
 - Gateway
-

5. Subnetting

Large network ko small networks me divide karna.

Advantages

- Better management
 - Reduced traffic
 - Better security
 - Efficient IP usage
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Subnet Formula

Number of Subnets

$$2^n$$

n = borrowed bits

Hosts per Subnet

$$2^h - 2$$

h = host bits

(-2 because network ID and broadcast ID)

Example

192.168.1.0/24

Agar /26 banaye:

Borrowed bits = 2

Subnets = $2^2 = 4$

Hosts = $2^6 - 2 = 62$

CIDR Notation

255.255.255.0 = /24

6. Supernetting

Small networks ko combine karna.

Purpose

- Routing table size reduce karna
- Faster routing

Example

200.1.0.0/24 + 200.1.1.0/24

Combined → 200.1.0.0/22

7. NAT (Network Address Translation)

Private IP ko Public IP me convert karta hai.

Purpose

- IPv4 addresses save karna
 - Security increase
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Types of NAT

1. Static NAT

One private ↔ one public IP

2. Dynamic NAT

Public IP pool use hota hai.

3. PAT (Port Address Translation)

One public IP + different ports.

Most common type.

8. IPv4

Features

- 32-bit addressing
- Dotted decimal notation

Example

192.168.1.1

IPv4 Classes

Class	Range
A	1–126
B	128–191
C	192–223
D	Multicast

Class	Range
E	Research

IPv4 Problems

- Limited addresses
- NAT dependency
- Less security

9. IPv6

IPv4 ka improved version.

Features

- 128-bit addressing
- Huge address space
- Better security
- Faster routing

Example

2001:db8::1

Total Addresses

2^{128}

IPv4 vs IPv6

Feature	IPv4	IPv6
Address Size	32-bit	128-bit
NAT	Needed	Not Needed
Security	Less	Better
Header	Complex	Simple

10. Routing

Routers routing table use karte hain.

Purpose

Source se destination tak best path select karna.

TYPES OF ROUTING

11. Static Routing

Routes manually configure kiye jaate hain.

Advantages

- Secure
- Simple

Disadvantages

- Large network me difficult
 - Automatic recovery nahi
-

12. Dynamic Routing

Routers automatically routes learn karte hain.

Advantages

- Scalable
- Automatic updates
- Fault tolerant

Disadvantages

- Complex
 - More overhead
-

Static vs Dynamic Routing

Feature	Static	Dynamic
Configuration	Manual	Automatic
Scalability	Low	High
Recovery	No	Yes

ROUTING ALGORITHMS

13. Shortest Path Algorithm

Shortest/least cost path find karta hai.

Metrics

- Hop count
- Delay
- Cost
- Bandwidth

Dijkstra Algorithm

Minimum cost path calculate karta hai.

Used In

OSPF

Example

$A \rightarrow B = 2$ $A \rightarrow C = 5$ $B \rightarrow C = 1$

Shortest Path: $A \rightarrow B \rightarrow C$

Cost: $2 + 1 = 3$

14. Flooding

Packet sab directions me send hota hai.

Advantages

- Reliable
- Simple

Disadvantages

- Huge traffic
 - Congestion
-

Control Methods

- Hop Count
 - Sequence Number
-

15. Distance Vector Routing

Har router distance table maintain karta hai.

Routers neighbors se routing information exchange karte hain.

Bellman-Ford Algorithm

Distance Vector Routing me use hota hai.

Advantages

- Simple

Disadvantages

- Slow convergence
 - Count to infinity problem
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Protocol Example

RIP

16. Link State Routing

Har router poore network ka topology map maintain karta hai.

Steps

1. Neighbor discovery
 2. Cost calculation
 3. Information flooding
 4. Database build
 5. Shortest path calculation
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Advantages

- Fast convergence
- Accurate routing

Disadvantages

- More memory needed
-

Protocol Example

OSPF

17. Hierarchical Routing

Large network ko regions me divide karna.

Advantages

- Smaller routing table
 - Better scalability
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18. Broadcast Routing

One sender → All devices

Example

ARP Request

19. Multicast Routing

One sender → Selected group

Example

Video streaming

Advantage

Bandwidth efficient

20. Anycast Routing

One sender → Nearest server/device

Example

DNS Servers CDN

Advantages

- Fast response
 - Load balancing
-

ROUTING PROTOCOLS

21. RIP (Routing Information Protocol)

Distance Vector Protocol.

Features

- Uses Hop Count
- Maximum Hop = 15
- Slow convergence

Advantages

- Simple

Disadvantages

- Small networks only
-

22. OSPF (Open Shortest Path First)

Link State Protocol.

Features

- Uses Dijkstra Algorithm
 - Fast convergence
 - Large network support
-

Advantages

- Efficient
- Fast routing

Disadvantages

- Complex
-

23. BGP (Border Gateway Protocol)

Internet backbone routing protocol.

Features

- Path Vector Protocol
- Policy based routing

Used By

ISPs and Internet

RIP vs OSPF vs BGP

Feature	RIP	OSPF	BGP
Type	Distance Vector	Link State	Path Vector
Algorithm	Bellman-Ford	Dijkstra	Path Vector
Metric	Hop Count	Cost	Path
Speed	Slow	Fast	Medium
Network Size	Small	Large	Internet

24. Fragmentation

Large packet ko small fragments me todna.

Occurs when: Packet > MTU

25. TTL (Time To Live)

Har router TTL ko 1 reduce karta hai.

TTL = 0 → Packet discard.

26. Convergence

Sab routers updated routing information par agree karte hain.

27. Routing Loop

Packet continuously network me circulate karta rehta hai.

28. Count to Infinity Problem

Distance Vector Routing me hota hai.

Solutions

- Split Horizon
 - Poison Reverse
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IMPORTANT NUMERICALS

Numerical 1

Hosts in /27 subnet.

Host bits: $32 - 27 = 5$

Hosts: $2^5 - 2 = 30$

Answer: 30 Hosts

Numerical 2

Subnets from /24 to /28.

Borrowed bits: 4

Subnets: $2^4 = 16$

Answer: 16 Subnets

IMPORTANT ONE-LINERS

Topic	Meaning
Routing	Best path selection
IP Address	Logical Address
Subnetting	Large → Small Networks
Supernetting	Small → Large Network
NAT	Private ↔ Public IP
IPv4	32-bit Addressing

Topic	Meaning
IPv6	128-bit Addressing
Flooding	Packet everywhere send
Distance Vector	Neighbor-based routing
Link State	Full topology routing
Broadcast	One → All
Multicast	One → Group
Anycast	One → Nearest
RIP	Distance Vector
OSPF	Link State
BGP	Internet routing
TTL	Packet lifetime
Fragmentation	Packet splitting

COMPLETE MIND MAP

NETWORK LAYER & ROUTING



